Hints: Functional sequences, series and spaces – 2. Series of functions

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Continuity is the M-test ($|x-r_n| \leqslant 1$). For differentiability, look at the difference quotient term by term: $\frac{|x+h-r_n|-|x-r_n|}{h}$ is bounded by 1 in absolute value, so the series of difference quotients is dominated by $\sum 2^{-n}$ uniformly in h.

Hint 2. At an irrational x each quotient tends to $\operatorname{sgn}(x-r_n)$ as $h \to 0$, so by dominated convergence for series (Tannery's theorem) $F'(x) = \sum_n 2^{-n} \operatorname{sgn}(x-r_n)$. So F is differentiable at every irrational point, in particular almost everywhere. (Alternatively: F is convex.)

Hint 3. At $x = r_k$, do the same with the k-th term separated: the right and left derivatives exist and differ by $2 \cdot 2^{-k}$, the jump of the derivative of $2^{-k}|x-r_k|$.

Problem 2

Hint 1. For each fixed $x \geqslant 0$ this is a Leibniz series, and the tail after N is at most $\frac{1}{N+1+x} \leqslant \frac{1}{N+1}$; non-absolute convergence is comparison with the harmonic series. The same tail bound gives $|S(x)| \leqslant \frac{1}{1+x} \to 0$.

Hint 2. For the asymptotics write $\frac{1}{n+x} = \int_0^1 t^{n+x-1}\,dt$ and sum the geometric series: $S(x) = \sum_{n\geqslant 1} \frac{(-1)^n}{n+x} = -\int_0^1 \frac{t^x}{1+t}\,dt$ (justify the interchange, e.g. by the uniform bound on the partial sums $\left|\sum_{n\leqslant N}(-t)^n\right| \leqslant 1$).

Hint 3. As $x \to \infty$ the mass of t^x concentrates at $t=1$, where $\frac{1}{1+t} = \frac{1}{2}$, and $\int_0^1 t^x\,dt = \frac{1}{x+1}$. Hence $S(x) \sim -\frac{1}{2x}$.

Problem 3

Hint 1. Represent $\frac{1}{(n+m)^2}$ as an integral that factors in n and m, e.g. $\frac{1}{(n+m)^2} = \int_0^\infty t e^{-(n+m)t}\,dt$.

Hint 2. The double sum becomes $\int_0^\infty t \left(\sum_{n=1}^N \frac{(-1)^n e^{-nt}}{n^n}\right)^2 dt$, which is $\geqslant 0$. It is strictly positive because the integrand is a continuous function that is not identically zero (a nonzero polynomial in e^{-t} has finitely many zeros).

Problem 4

Hint 1. Continuity: M-test with $c_n = a^n$. The term-by-term differentiated series is $-\sum (ab)^n \sin(b^n x)$; since $ab > 1$ it can converge only where $\sin(b^n x) \to 0$.

Hint 2. Show that $\sin(b^n x) \to 0$ cannot hold on a whole interval I: by the Riemann–Lebesgue lemma $\int_I \sin^2(b^n x)\,dx = \frac{1}{2}\int_I (1-\cos 2b^n x)\,dx \to \frac{|I|}{2}$, contradicting bounded convergence. So the differentiated series diverges on a dense set of points, and term-by-term differentiation is meaningless. Nowhere differentiability: Weierstrass's condition is b an odd integer and $ab > 1 + \frac{3\pi}{2}$ (proved in Differentiable functions – 8); Hardy showed that $ab \geqslant 1$ suffices.

Problem 5

Hint 1. The k-times differentiated series is $\sum(-\ln n)^k n^{-x}$. On $[1+\delta,\infty)$ it is dominated by $\sum(\ln n)^k n^{-1-\delta} < \infty$; use the M-test and the term-by-term differentiation theorem, then cover $(1,\infty)$ by such half-lines.

Hint 2. Non-uniformity on $(1,\infty)$: for any N and M, $\sup_{x>1}\sum_{n=N}^{M}\frac{\ln n}{n^x} = \sum_{n=N}^{M}\frac{\ln n}{n}$ (let $x\to 1^+$), which is unbounded in M. So the tails do not tend to 0 uniformly.

Hint 3. Compare with integrals: t^{-x} is decreasing, so $\frac{1}{x-1} = \int_1^\infty t^{-x}\,dt \leqslant \zeta(x) \leqslant 1+\int_1^\infty t^{-x}\,dt = 1+\frac{1}{x-1}$.

Problem 6

Hint 1. For $|x|\geqslant 1$ the terms do not tend to 0 (or are undefined, for $x=-1$ and odd n); for $|x|<1$ compare with the geometric series. On $[-a,a]$, $a<1$: $|\frac{x^n}{1+x^n}| \leqslant \frac{a^n}{1-a}$, so the M-test applies. The set is $(-1,1)$.

Hint 2. For $x\to 1^-$ put $x=e^{-t}$, $t\to 0^+$: the sum is $\sum_{n\geqslant 1}\frac{1}{e^{nt}+1}$, a Riemann sum with step t of a decreasing function. Hence it is $\sim \frac{1}{t}\int_0^\infty \frac{du}{e^u+1} = \frac{\ln 2}{t} \sim \frac{\ln 2}{1-x}$.

Problem 7

Hint 1. The partial sums S_n are continuous and increase pointwise to the continuous sum S, so $S-S_n$ decreases to 0: this is exactly Dini's theorem.

Hint 2. Counterexample: take any continuous $f_n\to 0$ pointwise but not uniformly (e.g. $f_n(x) = nxe^{-nx^2}$ on $[0,1]$) and telescope: $u_1=f_1$, $u_n=f_n-f_{n-1}$. The partial sums are f_n, the sum is 0.

Problem 8

Hint 1. Telescoping again turns a sequence example into a series example: find smooth $f_n\to f$ uniformly with f_n' converging pointwise to something other than f'.

Hint 2. Take $f_n(x)=\frac{x}{1+n^2x^2}$ from the standard zoo: $\sup|f_n| = \frac{1}{2n}$, $f_n'(x) = \frac{1-n^2x^2}{(1+n^2x^2)^2} \to \mathbf{1}_{\{0\}}(x)$. With $u_1=f_1$, $u_n=f_n-f_{n-1}$, the sum is 0, while the differentiated series sums to 1 at $x=0$.

Problem 9

Hint 1. Use the uniform Cauchy criterion: if $|S_n(x)-S_m(x)|\leqslant\varepsilon$ for all $x\in E$, then by continuity of S_n-S_m the same holds on $\overline{E}$. The rationals are dense in $[0,1]$.

Problem 10

Hint 1. Let $R_n = S-S_n$, the tails; they are continuous because S is. For fixed ε put $E_N = \{x : |R_n(x)|\leqslant\varepsilon \text{ for all } n\geqslant N\}$. Each E_N is closed and $\bigcup_N E_N = [0,1]$ by pointwise convergence.

Hint 2. By the Baire category theorem some E_N has non-empty interior, i.e. contains an interval I, on which $\sup_{n\geqslant N}|R_n|\leqslant\varepsilon$. Applying the same argument inside any given interval shows such intervals are dense. Repeating with $\varepsilon = 1, \frac{1}{2}, \frac{1}{4}, \ldots$ gives nested intervals, but they may shrink to a point, so uniform convergence on a subinterval does not follow.

Hint 3. For the example take the continuous $f_n\geqslant 0$ on $[0,1]$ from Hints 2–3 of Problem 10 in the homework of sheet 1: they converge pointwise to 0, but not uniformly on any subinterval. Telescoping, $u_1=f_1$, $u_n=f_n-f_{n-1}$, turns them into a series of continuous functions with partial sums f_n and continuous sum 0.